CSA1304 - THEORY OF COMPUTATION

LABORATORY EXPERIMENTS

Name: **Vimal Mahendran**

Reg No: **192311068**

1. Write a C program to simulate a Deterministic Finite Automata (DFA) for the given language representing strings that start with a and end with a

Program Code

```c
#include <stdio.h>
#include <string.h>

int main() {
    char s[100];
    int a,n;
    printf("Enter the string:");
    scanf("%s",s);
    n = strlen(s);
    if(s[0] == 'a' && s[n-1] == 'a') {
        printf("String accepted");
    }
    else {
        printf("String rejected");
    }
    return 0;
}
```

Output

```
Enter the string:abbababbab
String rejected
```

```
Enter the string:aababababbbaba
String accepted
```

2. Write a C program to simulate a Deterministic Finite Automata (DFA) for the given language representing strings that start with 0 and end with 1

Program code

```c
#include <stdio.h>
#include <string.h>

int main() {
    char s[100];
    int n;
    printf("Enter the string:");
    scanf("%s",s);
    n = strlen(s);
    if(s[0] == '0' && s[n-1] == '1') {
        printf("The string is valid");
    }
    else {
        printf("The string is invalid");
    }
    return 0;
}
```

Output

```
Enter the string:01010100110
The string is invalid
```

```
Enter the string:001010010001
The string is valid
```

3. Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)

S → 0A1 A → 0A | 1A | ε

Program Code

```c
#include <stdio.h>
#include <string.h>

int main() {
    char str[1000];
    printf("Enter the string:");
    if (scanf("%999s", str) != 1) return 1;
    int len = strlen(str);
    if (len < 2 || str[0] != '0' || str[len - 1] != '1') {
        printf("Rejected\n");
        return 0;
    }
    for (int i = 1; i < len - 1; i++) {
        if (str[i] != '0' && str[i] != '1') {
            printf("Rejected\n");
            return 0;
        }
    }
    printf("Accepted\n");
    return 0;
}
```

Output

```
Enter the string:01
Accepted
```

```
Enter the string:10
Rejected
```